Homework Assignment 3: Critical Systems by Diffusion, P_N and S_N

Tal Ramot

August 17, 2026

Question 1: Critical radius of a reflected sphere

Question 1 is the one *diffusion* question of this assignment — the S_N machinery set out before Question 3 belongs to Questions 3–5 — and the only one with two materials. A multiplying core of radius R_C is wrapped in $d \in \{1, 2, 3, 10\}$ mean free paths of reflector, and the cross sections are again the Question 5 table of Assignment 1.

The two regions

In both regions the substitution $u(r) = r\phi(r)$, which uses $\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\phi}{dr}\right) = \frac{1}{r}\frac{d^2u}{dr^2}$, turns the spherical diffusion equation into a plane one:

$$\frac{d^2u}{dr^2} - \frac{(1-c)\Sigma_t^2}{D_0}\,u = 0 \tag{1}$$

The core has $c_C > 1$, so the coefficient is negative and the solution oscillates; finiteness at the origin selects the sine. The reflector has $c_R \le 1$ and decays. With

$$B = k_0(c_C)\,\Sigma_{t,C}, \qquad \kappa = \frac{\Sigma_{t,R}}{\nu_0(c_R)}, \qquad R_{\text{ext}} = R_R + \frac{z_0(c_R)}{\Sigma_{t,R}} \tag{2}$$

the two profiles are

$$\phi_C(r) = A\,\frac{\sin(Br)}{r}, \qquad \phi_R(r) = F\,\frac{\sinh[\kappa\,(R_{\text{ext}} - r)]}{r} \tag{3}$$

k_0 and ν_0 are Case's discrete eigenvalue on its two branches, the same roots Assignment 1 used: $c\arctan(k_0) = k_0$ above $c = 1$ and $\operatorname{artanh}(1/\nu_0)/(1/\nu_0) = 1/c$ below it.

Matching, and the one criticality equation

The interface at $r = R_C$ carries a flux relation $\phi_R = g\,\phi_C$ and continuity of the net current $J = -D\,d\phi/dr$. Since $\phi = u/r$ gives $\phi'/\phi = u'/u - 1/r$, and the two logarithmic derivatives are $u_C'/u_C = B\cot(BR_C)$ and $u_R'/u_R = -\kappa\coth(\kappa L)$ with $L = R_{\text{ext}} - R_C$, all three approximations reduce to a single transcendental equation for R_C:

$$\boxed{D_C B\cot(BR_C) \;+\; g\,D_R\kappa\coth(\kappa L) \;-\; \frac{D_C - g\,D_R}{R_C} \;=\; 0} \tag{4}$$

with $\kappa L = [d + z_0(c_R)]/\nu_0(c_R)$, so the reflector enters only through its *optical* thickness. The last term has no slab counterpart: ϕ falls as $1/r$ even where u is flat, so a jump in D at a curved interface drives a current by itself. Equation (4) runs from $+\infty$ at $R_C \to 0$ to $-\infty$ at $R_C = \pi/B$, the unreflected limit, so the fundamental root is always bracketed by $(0, \pi/B)$ and `brentq` finds it in one call.

The three approximations differ only in what they put into D_0, z_0 and g:

	(a) continuous classic	(b) continuous asymptotic	(c) discontinuous asymptotic
$D_0(c)$	$1/3$	$\lvert 1-c\rvert\,\nu_0^2$	$\lvert 1-c\rvert\,\nu_0^2$
relaxation rate	$\sqrt{3\lvert 1-c\rvert}$	$k_0(c)$ or $1/\nu_0(c)$	$k_0(c)$ or $1/\nu_0(c)$
$z_0(c)$	$2/3$	$0.710446\,[1+q(1-c)^2]/c$	$0.710446\,[1+q(1-c)^2]/c$
$g = \phi_R/\phi_C$	1	1	$\mu_0(c_C)/\mu_0(c_R)$

Table 1: Only the last row makes (c) different from (b). Since $\mu_0 = 1/2$ for every medium in the classic theory, (a) is automatically continuous, and with no reflector at all (b) and (c) are the same bare sphere.

Zimmerman's μ_0. The discontinuity comes from Zimmerman (UCRL-82792), who writes the partial currents of the asymptotic mode as $\phi_\pm = \frac{1}{2}\mu_0\phi_0 \pm \frac{1}{2}\phi_1$, so that $\mu_0\phi_0 = J_+ + J_-$ is the total traffic across a surface and

$$\mu_0(c) = \frac{c\,\nu_0^2}{2}\ln\left(\frac{\nu_0^2}{\nu_0^2-1}\right) \xrightarrow[c>1]{} \frac{c}{2k_0^2}\ln\left(1+k_0^2\right) \tag{5}$$

the second form being the exact analytic continuation through $\nu_0 = i/k_0$. It is not an ansatz: integrating the discrete mode $\psi(\mu) \propto (\nu_0 - \mu)^{-1}$ over the two half-ranges reproduces it identically, and it returns the two textbook limits $\mu_0 = 1/2$ for an isotropic flux and $\mu_0 = 1$ for a one-way beam. Matching the traffic rather than the flux, $\mu_0(c_C)\phi_C = \mu_0(c_R)\phi_R$, is what makes ϕ itself jump. Nothing is lost at the interface — the same current crosses both faces.

material	c	mfp [cm]	k_0 or $1/\nu_0$	D_0	D [cm]	μ_0	z_0
Pu-239	1.5000	3.064	1.45110	0.23745	0.7275	0.40363	0.47127
U-235	1.3000	3.064	1.05708	0.26847	0.8225	0.43639	0.54552
H_2O	0.9000	3.064	0.52543	0.36222	1.1097	0.52660	0.78923
Fe	0.9980	4.300	0.07740	0.33387	1.4356	0.50050	0.71187
Na	1.0000	11.578	0.00000	0.33333	3.8595	0.50000	0.71045

Table 2: Asymptotic parameters of the five materials. Sodium is a pure scatterer, $\Sigma_c = \Sigma_f = 0$, so $c_R = 1$ exactly: ν_0 diverges, $\kappa = 0$, and the reflector solution degenerates to u_R linear in r with $\kappa\coth(\kappa L) \to 1/L$. The code takes that limit explicitly rather than dividing out a near-zero κ.

Results

The bare spheres of Assignment 1 Question 5, which every reflected radius below is measured against:

core	classic R_c [cm]	asympt. R_c [cm]	classic M_c [kg]	asympt. M_c [kg]
Pu-239	5.8163	5.1890	12.940	9.188
U-235	8.1031	7.4339	42.345	32.696

Table 3: Bare reference. Theory (a) is compared with the classical column, (b) and (c) with the asymptotic one.

reflector	d [mfp]	$R_c^{(a)}$	$R_c^{(b)}$	$R_c^{(c)}$	$M_c^{(a)}$	$M_c^{(b)}$	$M_c^{(c)}$
Water	1	5.3159	4.9484	4.5400	9.879	7.968	6.154
Water	2	5.0855	4.7987	4.3823	8.649	7.267	5.535
Water	3	5.0153	4.7499	4.3321	8.296	7.048	5.347
Water	10	4.9814	4.7244	4.3061	8.129	6.935	5.251
Iron	1	5.4212	4.9575	4.6450	10.478	8.013	6.591
Iron	2	5.1056	4.7364	4.4182	8.753	6.988	5.672
Iron	3	4.9506	4.6237	4.3059	7.979	6.501	5.250
Iron	10	4.6790	4.4188	4.1070	6.737	5.674	4.556
Sodium	1	6.4397	5.7159	5.5171	17.562	12.281	11.044
Sodium	2	6.3134	5.6405	5.4318	16.549	11.801	10.540
Sodium	3	6.2505	5.6022	5.3892	16.059	11.563	10.293
Sodium	10	6.1315	5.5286	5.3080	15.159	11.113	9.835

Table 4: Pu-239 core: critical radius in cm and critical mass in kg, under (a) continuous classic, (b) continuous asymptotic and (c) discontinuous asymptotic diffusion. Bare: 5.8163 cm / 12.940 kg classic, 5.1890 cm / 9.188 kg asymptotic.

reflector	d [mfp]	$R_c^{(a)}$	$R_c^{(b)}$	$R_c^{(c)}$	$M_c^{(a)}$	$M_c^{(b)}$	$M_c^{(c)}$
Water	1	7.2501	6.9113	6.5228	30.330	26.273	22.087
Water	2	6.9185	6.6731	6.2694	26.355	23.649	19.611
Water	3	6.8146	6.5937	6.1865	25.186	22.816	18.844
Water	10	6.7638	6.5520	6.1432	24.627	22.385	18.451
Iron	1	7.2811	6.8392	6.5656	30.720	25.460	22.525
Iron	2	6.8002	6.4625	6.1789	25.027	21.480	18.775
Iron	3	6.5545	6.2644	5.9804	22.411	19.565	17.023
Iron	10	6.1125	5.8970	5.6199	18.176	16.320	14.126
Sodium	1	8.4264	7.7951	7.6194	47.618	37.697	35.205
Sodium	2	8.2322	7.6564	7.4692	44.401	35.721	33.164
Sodium	3	8.1328	7.5844	7.3919	42.811	34.722	32.145
Sodium	10	7.9401	7.4425	7.2414	39.840	32.809	30.221

Table 5: U-235 core, same columns. Bare: 8.1031 cm / 42.345 kg classic, 7.4339 cm / 32.696 kg asymptotic.

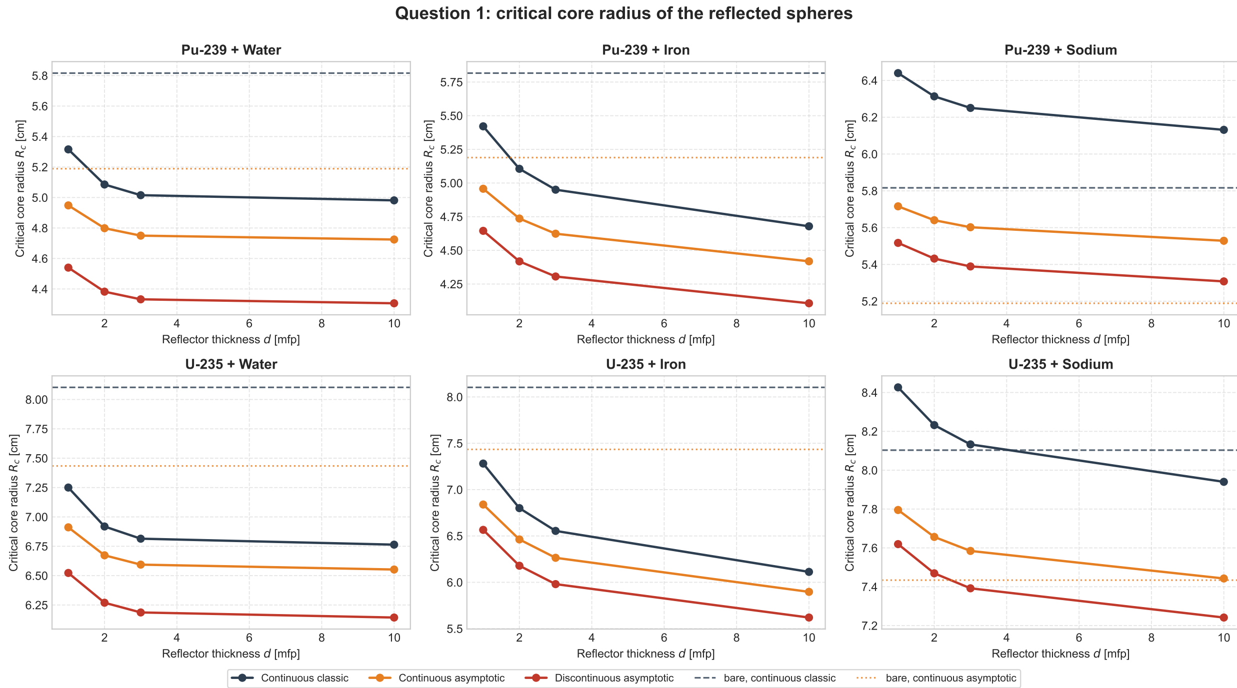

Figure 1: Critical core radius against reflector optical thickness. Dashed and dotted lines are the two bare references. Water saturates by $d \approx 3$ mfp; iron is still improving at $d = 10$; sodium sits above its bare line throughout.

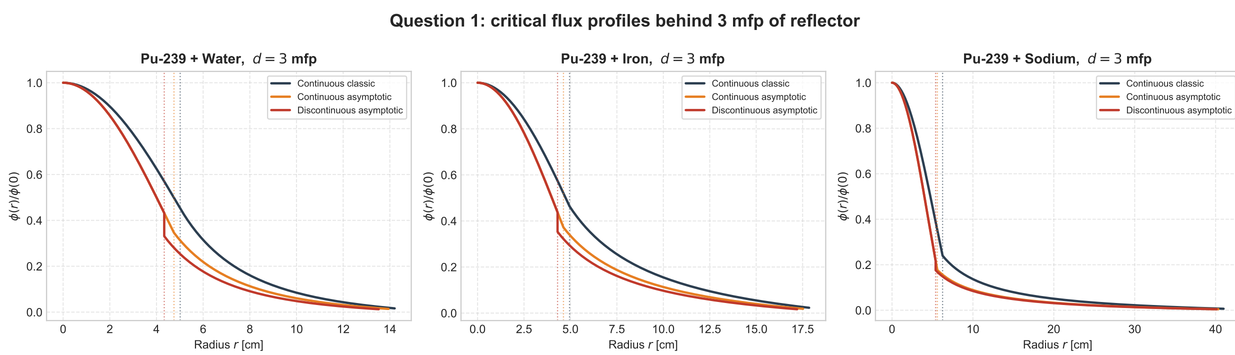

Figure 2: Critical flux across core and reflector for the Pu-239 core behind 3 mfp, normalised to $\phi(0) = 1$; dotted verticals mark each theory's interface. The discontinuous curve drops by the factor g at the interface and is continuous in current, not in flux. Note the horizontal scale of the sodium panel: 3 mfp of sodium is 34.7 cm.

What the three approximations say

The reflector saving. Water and iron behave as reflectors should. Both cores lose 12–17% of their radius behind water and 12–24% behind iron by theory (c), which because the mass goes as R^3 is a factor of nearly two in critical mass: Pu-239 falls from 9.188 kg bare to 4.556 kg behind 10 mfp of iron. The two reflectors saturate differently, and the reason is c_R. Water absorbs ($c_R = 0.90$, $\nu_0 = 1.90$), so a neutron more than about two mean free paths deep never returns and the water columns are flat past $d = 3$. Iron barely absorbs ($c_R = 0.998$, $\nu_0 = 12.9$), so its return keeps improving out to $d = 10$ and beyond — iron starts as the worse reflector at $d = 1$ and ends the better one at $d = 10$.

The ordering of the three theories. At every entry $R_c^{(a)} > R_c^{(b)} > R_c^{(c)}$. The first gap is inherited from the bare problem: the asymptotic $D_0(c_C)$ and $k_0(c_C)$ already give a smaller sphere than $D_0 = 1/3$ does, by 11% for Pu-239. The second is the jump itself. Every pair here has $\mu_0(c_C) < \mu_0(c_R)$, so $g < 1$ — 0.766 for Pu-239 behind water, 0.873 for U-235 behind iron — because the multiplying core has the shorter relaxation length and hence the more forward-peaked angular flux, and so carries more current per unit density. A g below one multiplies down both the leakage term and the curvature term of (4), which is why (c) always predicts the smallest core, by a further 8–9% for water and iron.

The sodium columns are diffusion theory failing. Every sodium radius above is *larger* than the corresponding bare sphere, which no reflector can do. The condition is being solved correctly — (4) was checked against an independent two-region spherical finite-volume k-eigenvalue solve, 3000 cells with harmonic-mean face diffusion coefficients across the material jump, and the two agree to better than 0.4% on exactly these cases. The fault is in the model. Splitting the interface leakage of the Pu-239 core at $d = 10$ mfp into its two parts, $J/\phi = D_R/R_c + D_R/L$:

reflector	D_R [cm]	shell [cm]	geometric D_R/R_c	far-face D_R/L	total vs. bare 0.5592
Water	1.021	30.6	0.2050	0.1826	0.3876
Iron	1.433	43.0	0.3063	0.0381	0.3444
Sodium	3.859	115.8	**0.6295**	0.0312	0.6607

Table 6: Leakage ratio at the interface, in cm^{-1}, classic theory. Sodium's far-face return is the best of the three — 10 mfp of a pure scatterer really does send almost everything back — but its geometric term alone exceeds what a bare vacuum surface leaks.

The geometric term is the current driven by the $1/r$ spreading of $\phi = u/r$, and it is harmless only while $D_R \ll R_c$. Sodium's $\Sigma_t = 0.086368\ \text{cm}^{-1}$ is a mean free path of 11.58 cm against a core radius near 6 cm: a neutron entering the reflector travels two core radii before its first collision, which is precisely the regime Fick's law expands away from. None of the three approximations repairs it, because a pure scatterer has $D_0 = 1/3$ in all of them and $D_R = 3.86$ cm throughout. Measured against its own bare sphere at $d = 10$, (a) is +5.4%, (b) +6.6% and (c) +2.3% — only the Zimmerman jump helps, and only because $g = 0.807$ multiplies the offending term. Transport theory would not make this mistake: the shell is optically thick and purely scattering, so its true albedo is high and the true critical radius must lie below the bare 5.1890 cm.

Question 2: Critical slab half-thickness by P_N

A homogeneous slab of total thickness $2a$, reflective at the midplane $x = 0$ and vacuum at $x = a$, in mean free paths ($\Sigma_t = 1$), for $c \in \{1.2, 1.5, 1.8\}$ and $N \in \{1, 3, 5, 9\}$. The unknown is the half-thickness a_{crit} at which $k_{\text{eff}} = 1$.

A notational warning. This section's a is the *half*-thickness, so it is what Question 3 calls $a/2$. The two tables are comparable entry by entry only after that identification.

The P_N equations

With isotropic scattering and fission lumped as in the paragraph "The meaning of c" before Question 3, and the fission source divided by k,

$$\mu \frac{\partial \psi}{\partial x} + \psi(x, \mu) = \frac{c}{2k}\,\phi_0(x), \qquad \phi_n(x) = \int_{-1}^{1} P_n(\mu)\,\psi(x,\mu)\,d\mu \tag{6}$$

Expanding $\psi(x, \mu) = \sum_{n \geq 0} \frac{2n+1}{2}\phi_n(x) P_n(\mu)$, multiplying (6) by $P_n(\mu)$ and integrating over μ turns the streaming term into a three-term coupling through the Legendre recurrence $(2n+1)\mu P_n = (n+1)P_{n+1} + nP_{n-1}$, while the collision term is diagonal and the isotropic source reaches only $n = 0$:

$$\frac{n+1}{2n+1}\frac{d\phi_{n+1}}{dx} + \frac{n}{2n+1}\frac{d\phi_{n-1}}{dx} + \phi_n(x) = \delta_{n0}\,\frac{c}{k}\,\phi_0(x), \qquad n = 0, \ldots, N \tag{7}$$

The system closes on itself once $\phi_{N+1} \equiv 0$ is imposed, which is the entire content of the P_N approximation. In vector form, with $\mathbf{\Phi} = (\phi_0, \ldots, \phi_N)^T$,

$$\tilde{\mathbf{A}}\,\frac{d\mathbf{\Phi}}{dx} + \mathbf{\Sigma}(k)\,\mathbf{\Phi} = 0, \qquad \tilde{A}_{n,n+1} = \frac{n+1}{2n+1}, \quad \tilde{A}_{n,n-1} = \frac{n}{2n+1}, \qquad \mathbf{\Sigma}(k) = \mathbf{I} - \frac{c}{k}\,\mathbf{e}_0\mathbf{e}_0^T \tag{8}$$

$\tilde{\mathbf{A}}$ is the Jacobi matrix of the Legendre recurrence truncated at order N, so its $N+1$ eigenvalues are exactly the roots of $P_{N+1}(\mu)$ — the Gauss–Legendre ordinates of S_{N+1}. That is the algebraic content of the claim in Question 3 that the two methods share their interior equations and differ only at the boundary.

Boundary conditions

At the midplane the flux is even in μ, so every odd moment vanishes:

$$\phi_{2m+1}(0) = 0, \qquad m = 0, \ldots, \tfrac{N-1}{2} \tag{9}$$

Equivalently $d\phi_{2m}/dx|_0 = 0$, which (7) then delivers for free; only one of the two families may be imposed or the problem is over-determined.

At $x = a$ no neutron enters, $\psi(a, \mu) = 0$ for $\mu < 0$, which $N+1$ moments cannot satisfy pointwise. Marshak's choice is to annihilate the $\frac{N+1}{2}$ lowest odd moments of the incoming half-range,

$$\int_{-1}^{0} \mu^{2m-1}\psi(a, \mu)\,d\mu = 0 \implies \sum_{n=0}^{N} c_{mn}\,\phi_n(a) = 0, \qquad c_{mn} = \frac{2n+1}{2}\int_{-1}^{0} \mu^{2m-1} P_n(\mu)\,d\mu \tag{10}$$

for $m = 1, \ldots, \frac{N+1}{2}$. Weighting with $P_{2m-1}(\mu)$ instead of μ^{2m-1} gives the same set of conditions, since the odd Legendre polynomials up to degree N and the odd monomials up to degree N span the same space; the two differ by an invertible mixing of the rows and have the same null space. The coefficients follow from $P_n(-\mu) = (-1)^n P_n(\mu)$,

$$c_{mn} = (-1)^{n+1}\,\frac{2n+1}{2}\int_0^1 \mu^{2m-1} P_n(\mu)\,d\mu \tag{11}$$

whose integrand is a polynomial of degree at most $2N$, so an $(N+1)$-point Gauss–Legendre rule mapped to $[0,1]$ evaluates it exactly rather than approximately.

Counting: (7) is $N+1$ first-order equations and needs $N+1$ conditions; (9) supplies $\frac{N+1}{2}$ and (10) the other $\frac{N+1}{2}$. For $N=1$, (10) reads $-\frac{1}{4}\phi_0(a) + \frac{1}{2}\phi_1(a) = 0$, i.e. $\phi_0(a) = 2\phi_1(a)$: the classical statement that the scalar flux at a vacuum surface is twice the net current, and the origin of the extrapolation length $2/3$ quoted in Question 3.

Method 1: box discretisation and the Bell & Glasstone k iteration

The scheme. Divide $[0,a]$ into J cells of width $\Delta x = a/J$ with nodes $x_j = j\Delta x$. Integrating (8) over one cell and closing it with the midpoint rule — derivative by the difference across the cell, value by the average of the two faces —

$$\left.\frac{d\phi_n}{dx}\right|_{j+\frac{1}{2}} \approx \frac{\phi_{n,j+1} - \phi_{n,j}}{\Delta x}, \qquad \phi_{n,j+\frac{1}{2}} \approx \frac{\phi_{n,j+1} + \phi_{n,j}}{2} \tag{12}$$

gives one $(N+1)$-row block per cell,

$$\left(\frac{\tilde{\mathbf{A}}}{\Delta x} + \frac{\mathbf{I}}{2}\right)\boldsymbol{\Phi}_{j+1} + \left(\frac{\mathbf{I}}{2} - \frac{\tilde{\mathbf{A}}}{\Delta x}\right)\boldsymbol{\Phi}_j = q_{j+\frac{1}{2}}\,\mathbf{e}_0, \qquad j = 0,\ldots,J-1 \tag{13}$$

This is the Keller box scheme, and it is the diamond difference of (27) written in moments rather than in ordinates: the same closure, applied to $N+1$ coupled unknowns per node instead of one per ordinate. Two consequences of that coupling: the sweep of Question 3 is not available, because $\tilde{\mathbf{A}}$ mixes moments that stream in both directions, so the cells must be solved simultaneously; and there is no negative-flux fixup, because ϕ_n for $n \geq 1$ is legitimately of either sign and a sign change carries no information.

The linear system. Stacking the $J+1$ nodal vectors gives $(J+1)(N+1)$ unknowns. Equation (13) supplies $J(N+1)$ rows, (9) supplies $\frac{N+1}{2}$ rows acting on $\boldsymbol{\Phi}_0$ and (10) the last $\frac{N+1}{2}$ acting on $\boldsymbol{\Phi}_J$; the count closes exactly. Ordered node by node the matrix is block-bidiagonal with two boundary blocks, i.e. banded with half-bandwidth $N+1$, and one banded (or sparse) LU factorisation solves it. Call that matrix $\mathbf{L}$: it carries streaming, collision and both boundary conditions, and it does *not* depend on k.

Why the source must stay on the right. The c/k term cannot be absorbed into $\mathbf{L}$. If it were, the system would be homogeneous and would admit a non-trivial solution only at the critical k — which is the answer, not a way of computing it. Left on the right-hand side it becomes a fission source lagged one iteration,

$$\mathbf{L}\,\boldsymbol{\Phi}^{(\ell+1)} = \frac{c}{k^{(\ell)}}\,\mathbf{F}\,\boldsymbol{\Phi}^{(\ell)}, \qquad q^{(\ell)}_{j+\frac{1}{2}} = \frac{c}{k^{(\ell)}}\cdot\frac{\phi^{(\ell)}_{0,j+1} + \phi^{(\ell)}_{0,j}}{2} \tag{14}$$

which is the power method on $\mathbf{L}^{-1}\mathbf{F}$, whose dominant eigenvalue is k_{eff}. Because c lumps scattering and fission into one isotropic term, there is no separate scattering source and hence no inner iteration at all: one factorisation, then one back-substitution per outer step, against the ten sweeps per outer that Question 5 needs.

The k update. Bell & Glasstone update k by the ratio of successive fission productions,

$$k^{(\ell+1)} = k^{(\ell)} \frac{\displaystyle\int_0^a \phi_0^{(\ell+1)}\,dx}{\displaystyle\int_0^a \phi_0^{(\ell)}\,dx} \tag{15}$$

the constant c cancelling between numerator and denominator. The integral is taken as $\Delta x \sum_j \phi_{0,j+1/2}$, which is the trapezoid rule on the nodes; this matters, because it is the same midpoint average the source (13) uses, so (15) is exactly the Rayleigh ratio of the discrete operator and not a second, inconsistent discretisation of the same integral. The iterate is then rescaled by $\max_j \phi_{0,j}$, which has no effect on (15) — it only keeps the vector from drifting towards 0 or ∞ as k^ℓ compounds — and the loop stops on $|k^{(\ell+1)} - k^{(\ell)}| < \epsilon_k$. Convergence is linear at the dominance ratio of the two leading eigenvalues, which for a bare slab is well separated, and the starting guess

$$\phi_0^{(0)}(x) = \cos\!\left(\frac{\pi x}{2\,(a + 0.7104)}\right), \qquad k^{(0)} = 1 \tag{16}$$

is the diffusion fundamental mode with the asymptotic extrapolation distance, already almost the answer.

The outer loop. Dividing the fission source by k makes a solution exist at every thickness, so criticality is a root search on

$$f(a) = k(a) - 1 \tag{17}$$

exactly as in Questions 3–5. $k(a)$ is monotone increasing — a thicker slab leaks a smaller fraction — and runs from $k \to 0$ as $a \to 0$ to the infinite-medium value $k \to c$ as $a \to \infty$. So for every $c > 1$ a root exists and is unique, and any bracket $[a_{\text{lo}}, a_{\text{hi}}]$ straddling the diffusion estimate below is admissible; `brentq` closes it in a handful of evaluations, each of which is one full power iteration.

Errors. The box scheme is second order, so the discretisation error in a_{crit} falls as Δx^2 and is separable from the truncation error in N by running the same N on a refined mesh — the check reported in Question 3 for S_N, and the quantity that Method 2 removes altogether.

Method 2: modal elimination, as an N-exact benchmark

Method 1 carries two errors, one in N and one in Δx. Method 2 removes the second by solving (8) in closed form at $k = 1$ and imposing criticality directly on the boundary conditions.

Split the $N+1$ moments into $M = \frac{N+1}{2}$ even and M odd, $\mathbf{\Phi}_{\text{e}} = (\phi_0, \phi_2, \dots)$ and $\mathbf{\Phi}_{\text{o}} = (\phi_1, \phi_3, \dots)$. Every derivative in (7) changes the parity of its index, so the even-n and odd-n equations separate:

$$\mathbf{A}\frac{d\mathbf{\Phi}_{\text{o}}}{dx} + \mathbf{\Sigma}_0(k)\,\mathbf{\Phi}_{\text{e}} = 0, \qquad \mathbf{B}\frac{d\mathbf{\Phi}_{\text{e}}}{dx} + \mathbf{\Phi}_{\text{o}} = 0, \qquad \mathbf{\Sigma}_0(k) = \operatorname{diag}\!\big(1 - \tfrac{c}{k}, 1, \dots, 1\big) \tag{18}$$

$\mathbf{A}$ and $\mathbf{B}$ being the two off-diagonal blocks of $\tilde{\mathbf{A}}$ under that reordering, $\tilde{\mathbf{A}} = \left(\begin{smallmatrix} 0 & \mathbf{A} \\ \mathbf{B} & 0 \end{smallmatrix}\right)$, with rows $A_{m,\cdot}$ from $n = 2m$ and $B_{m,\cdot}$ from $n = 2m+1$. Substituting $\mathbf{\Phi}_\mathrm{o} = -\mathbf{B}\,\mathbf{\Phi}_\mathrm{e}'$ into the first equation eliminates the odd moments and leaves a coupled Helmholtz system,

$$\frac{d^2\mathbf{\Phi}_\mathrm{e}}{dx^2} + \mathbf{K}^2(k)\,\mathbf{\Phi}_\mathrm{e} = 0, \qquad \mathbf{K}^2(k) = -\,(\mathbf{AB})^{-1}\mathbf{\Sigma}_0(k) \tag{19}$$

The minus sign is not cosmetic and is the easiest thing to lose: the elimination produces $\mathbf{AB}\,\mathbf{\Phi}_\mathrm{e}'' = \mathbf{\Sigma}_0\mathbf{\Phi}_\mathrm{e}$, and moving that to the form (19) flips it. At $N = 1$ it reads $\phi_0'' + 3(c-1)\phi_0 = 0$, the classic diffusion buckling $B = \sqrt{3(c-1)}$ of the table in Question 1 — with the sign dropped one gets cosh where cos belongs and no critical thickness at all.

The spectrum of $\mathbf{K}^2$. Since $\tilde{\mathbf{A}}^2 = \left(\begin{smallmatrix} \mathbf{AB} & 0 \\ 0 & \mathbf{BA} \end{smallmatrix}\right)$ and the eigenvalues of $\tilde{\mathbf{A}}$ are the $N+1$ roots $\pm\mu_i$ of P_{N+1}, the matrix $\mathbf{AB}$ has the M eigenvalues $\mu_i^2 > 0$ and is similar to a positive definite matrix. By Sylvester's law of inertia the eigenvalues of (19) then carry the signs of $-\mathbf{\Sigma}_0$, which for $c > 1$ at $k = 1$ has one negative and $M-1$ positive entries: **exactly one eigenvalue $\lambda_1^2 = B_0^2 > 0$, and $M-1$ eigenvalues $\lambda_j^2 = -\kappa_j^2 < 0$.** The one oscillatory mode is the fundamental; the rest are the boundary layers, and at $c = 1$ the positive one collapses to $B_0 = 0$, an infinite critical slab.

Assembling the criticality condition. Diagonalise $\mathbf{K}^2(1) = \mathbf{V\Lambda V}^{-1}$. The solution even about $x = 0$, which satisfies (9) identically because $\mathbf{\Phi}_\mathrm{o} = -\mathbf{B}\mathbf{\Phi}_\mathrm{e}'$ vanishes wherever $\mathbf{\Phi}_\mathrm{e}'$ does, is

$$\mathbf{\Phi}_\mathrm{e}(x) = C_1\mathbf{v}_1\cos(B_0x) + \sum_{j=2}^{M} C_j\mathbf{v}_j\cosh(\kappa_jx) \tag{20}$$

$$\mathbf{\Phi}_\mathrm{o}(x) = C_1(\mathbf{Bv}_1)B_0\sin(B_0x) - \sum_{j=2}^{M} C_j(\mathbf{Bv}_j)\kappa_j\sinh(\kappa_jx) \tag{21}$$

M free constants against the M Marshak conditions (10). Splitting the coefficient matrix c_{mn} into its even and odd columns $\mathbf{M}_\mathrm{e}, \mathbf{M}_\mathrm{o}$ and evaluating at $x = a$ gives M homogeneous equations $\mathbf{H}(a)\mathbf{C} = \mathbf{0}$ with

$$\mathbf{H}_{\cdot 1} = \mathbf{M}_\mathrm{e}\mathbf{v}_1\cos(B_0a) + \mathbf{M}_\mathrm{o}(\mathbf{Bv}_1)B_0\sin(B_0a), \quad \mathbf{H}_{\cdot j} = \mathbf{M}_\mathrm{e}\mathbf{v}_j\cosh(\kappa_ja) - \mathbf{M}_\mathrm{o}(\mathbf{Bv}_j)\kappa_j\sinh(\kappa_ja) \tag{22}$$

and a non-trivial flux exists only where

$$\boxed{\det\mathbf{H}(a) = 0} \tag{23}$$

a_crit is its smallest positive root. Two numerical points: the cosh and sinh columns overflow once $\kappa_ja \gtrsim 700$, so each is divided by its own $\cosh(\kappa_ja)$, which leaves $\mathbf{v}_j$ and $-(\mathbf{Bv}_j)\kappa_j\tanh(\kappa_ja)$ and cannot move a zero of the determinant, a column scaling being a non-zero factor of it; and the determinant itself spans many orders of magnitude across the bracket, so it is the sign change that is tracked, not the value.

The two methods against P_1 in closed form

At $N = 1$ everything above is elementary and both methods must reproduce it. With $B = \sqrt{3(c-1)}$ and $\phi_0 = \cos(Bx)$, the Marshak condition $\phi_0(a) = 2\phi_1(a)$ and $\phi_1 = -\frac{1}{3}\phi_0'$ give $\cot(Ba) = \frac{2}{3}B$, that is

$$a_\mathrm{crit}^{P_1} = \frac{1}{B}\arctan\left(\frac{3}{2B}\right) \tag{24}$$

c	$B = \sqrt{3(c-1)}$	P_1 Marshak	P_1 Mark	S_2 of Q3	exact transport
1.2	0.774597	1.412499	1.484982	1.48499	1.28981
1.5	1.224745	0.723479	0.780013	0.78001	0.60762
1.8	1.549193	0.496559	0.542907	0.54291	0.39314

Table 7: Analytic P_1 half-thickness in mean free paths, against Question 3's S_2 column (there tabulated as $a/2$). The Mark column, $\arctan(\sqrt{3}/B)/B$, is the same equation with the extrapolation length $1/\sqrt{3}$ in place of $2/3$, and it reproduces the S_2 entry to five digits — the demonstration promised in Question 3 that P_N and S_{N+1} differ in slab geometry by their boundary condition alone. Both are the $N = 1$ target that Method 1 must approach as $\Delta x \to 0$ and that Method 2 must return outright.

The S_N method of Questions 3–5

All three questions solve the same monoenergetic, steady-state problem,

$$\mu \frac{\partial \psi}{\partial x} + \Sigma_t \psi = \frac{1}{2}\left[\Sigma_s \phi + \frac{1}{k}\nu \Sigma_f \phi\right], \qquad \phi(x) = \int_{-1}^{1} \psi(x,\mu)\, d\mu \tag{25}$$

with the same three ingredients: a Gauss–Legendre quadrature in angle, diamond differencing in space with a negative-flux fixup, and the Bell & Glasstone k iteration. Lengths are mean free paths in Questions 3 and 4 and centimetres in Question 5.

Angle. The N ordinates $\{\mu_m\}$ and weights $\{w_m\}$ are the Gauss–Legendre nodes on $[-1, 1]$, so that $\mu_{N-m+1} = -\mu_m$, $\sum_m w_m = 2$, and

$$\phi_i = \sum_{m=1}^{N} w_m \psi_{i,m} \tag{26}$$

Space, slab. Integrating over cell i and closing with the diamond relation $\psi_{i,m} = \frac{1}{2}(\psi_{i+1/2,m} + \psi_{i-1/2,m})$ gives the sweep

$$\psi_{i,m} = \frac{2|\mu_m|\, \psi_{\text{in},m} + \frac{1}{2} S_i \Delta x}{\Sigma_t \Delta x + 2|\mu_m|}, \qquad \psi_{\text{out},m} = 2\psi_{i,m} - \psi_{\text{in},m} \tag{27}$$

swept from $x = a/2$ inwards for $\mu_m < 0$ and from $x = 0$ outwards for $\mu_m > 0$.

Space, sphere. The spherical equation carries an angular redistribution term,

$$\mu \frac{\partial \psi}{\partial r} + \frac{1-\mu^2}{r}\frac{\partial \psi}{\partial \mu} + \Sigma_t \psi = \frac{S}{2} \tag{28}$$

whose conservative form integrates over the cell to

$$\mu_m \left(A_{i+\frac{1}{2}} \psi_{i+\frac{1}{2},m} - A_{i-\frac{1}{2}} \psi_{i-\frac{1}{2},m}\right) + \frac{A_{i+\frac{1}{2}} - A_{i-\frac{1}{2}}}{w_m}\left(\alpha_{m+\frac{1}{2}} \psi_{i,m+\frac{1}{2}} - \alpha_{m-\frac{1}{2}} \psi_{i,m-\frac{1}{2}}\right) + \Sigma_t V_i \psi_{i,m} = \frac{S_i}{2} V_i \tag{29}$$

with $A = 4\pi r^2$ and $V_i = \frac{4\pi}{3}(r_{i+1/2}^3 - r_{i-1/2}^3)$. This is (27) with a second outgoing face, in angle, closed by the same diamond relation $\psi_{i,m} = \frac{1}{2}(\psi_{i,m+1/2} + \psi_{i,m-1/2})$. The angular coefficients follow from demanding that a flat flux solve (29) exactly:

$$\alpha_{\frac{1}{2}} = 0, \qquad \alpha_{m+\frac{1}{2}} = \alpha_{m-\frac{1}{2}} - w_m\mu_m, \qquad \alpha_{N+\frac{1}{2}} = 0 \;\text{ since } \sum_m w_m\mu_m = 0 \tag{30}$$

The recursion needs a starting value $\psi_{i,1/2}$ at $\mu_{1/2} = -1$, where $1-\mu^2 = 0$ and the angular derivative drops out; what remains, $-\partial\psi/\partial r + \Sigma_t\psi = S/2$, is the plain slab sweep at $|\mu| = 1$, run inwards from the vacuum boundary.

One cell solve for both geometries.

Equations (27) and (29) are the same equation over a different number of faces, and it is worth writing that form down once because it is what the code implements. Let a cell carry a list of outgoing faces f, each contributing $a^f_{\text{out}}\psi^f_{\text{out}} - a^f_{\text{in}}\psi^f_{\text{in}}$ to the balance:

$$\sum_f \left(a^f_{\text{out}}\psi^f_{\text{out}} - a^f_{\text{in}}\psi^f_{\text{in}}\right) + \Sigma_t V\psi = \tfrac{1}{2}S\,V \tag{31}$$

Closing *every* face with the same diamond relation $\psi^f_{\text{out}} = 2\psi - \psi^f_{\text{in}}$ and collecting ψ gives one formula for any geometry:

$$\psi = \frac{\frac{1}{2}SV + \sum_f \left(a^f_{\text{out}} + a^f_{\text{in}}\right)\psi^f_{\text{in}}}{\Sigma_t V + 2\sum_f a^f_{\text{out}}} \tag{32}$$

The two geometries differ only in what they put in the list:

	faces	a_{out}	a_{in}
slab	space	$\lvert\mu_m\rvert$	$\lvert\mu_m\rvert$
sphere	space	$\lvert\mu_m\rvert A_{\text{out}}$	$\lvert\mu_m\rvert A_{\text{in}}$
	angle	$\frac{A_{i+1/2}-A_{i-1/2}}{w_m}\,\alpha_{m+\frac{1}{2}}$	$\frac{A_{i+1/2}-A_{i-1/2}}{w_m}\,\alpha_{m-\frac{1}{2}}$

with $V = \Delta x$ in the slab. Substituting the slab row into (32) returns (27) exactly, and the sphere's two rows return (29).

Fixup. Whenever a diamond outgoing flux $2\psi_{i,m} - \psi_{\text{in},m}$ comes out negative it is set to zero and the cell balance is re-solved with it held there, in angle as well as in space. This never occurs in Questions 3–5: the fission source is proportional to the flux itself, so no cell is ever starved, and a critical system is thin enough that $\Sigma_t\Delta x/|\mu_m| < 2$ at every ordinate.

Boundary conditions. Reflective at the symmetry point, $\psi_{1/2,m} = \psi_{1/2,m'}$ with $\mu_{m'} = -\mu_m$; vacuum at the outer face, $\psi_{I+1/2,m} = 0$ for $\mu_m < 0$.

Eigenvalue. Outer iteration l fixes the fission source $Q_{f,i}^{(l)} = \nu\Sigma_{f,i}\phi_i^{(l)}/k^{(l)}$; inner iteration n sweeps $S_i = \Sigma_{s,i}\phi_i^{(n)} + Q_{f,i}^{(l)}$ until ϕ settles; then

$$k^{(l+1)} = k^{(l)}\,\frac{P^{(l+1)}}{P^{(l)}}, \qquad P^{(l)} = \sum_i \nu\Sigma_{f,i}\,\phi_i^{(l)}\,V_i \tag{33}$$

Dividing the fission source by k makes a solution exist at *every* system size, so criticality becomes a root search: `brentq` on $k(\text{size}) - 1$, which costs 8 evaluations of k.

The meaning of c. Questions 3 and 4 specify only $c = (\Sigma_s + \nu\Sigma_f)/\Sigma_t$. At $k = 1$ the source bracket collapses to $c\,\Sigma_t\phi$ whatever the split between scattering and fission, so the critical size depends on c alone and the code takes the simplest split, $\Sigma_t = 1$, $\Sigma_s = 0$, $\nu\Sigma_f = c$. Question 5 has no such freedom and uses the tabulated cross sections directly.

Question 3: Critical slab half-thickness by S_N

Reflective at $x = 0$, vacuum at $x = a/2$, 200 cells. The reference column is Assignment 1's exact-transport relation $a/2 = \frac{\pi}{2}\nu_0 - z_0$, evaluated from Case's tabulated ν_0 and z_0.

c	order	$a/2$ [mfp]	exact transport [mfp]	departure
1.2	S_2	1.48499	1.28981	+15.13%
1.2	S_4	1.31855	1.28981	+2.23%
1.2	S_6	1.29861	1.28981	+0.68%
1.2	S_{10}	1.29218	1.28981	+0.18%
1.5	S_2	0.78001	0.60762	+28.37%
1.5	S_4	0.64949	0.60762	+6.89%
1.5	S_6	0.62089	0.60762	+2.18%
1.5	S_{10}	0.60904	0.60762	+0.23%
1.8	S_2	0.54291	0.39314	+38.09%
1.8	S_4	0.43875	0.39314	+11.60%
1.8	S_6	0.41014	0.39314	+4.32%
1.8	S_{10}	0.39459	0.39314	+0.37%

Table 8: Critical half-thickness in mean free paths. The departure is entirely angular: the same radii are reproduced to seven digits on meshes from 50 to 800 cells.

The exact one-speed value at $c = 1.5$ is 0.605055 mfp, the Sood *et al.* Pu-239 slab (1.853722 cm at $\Sigma_t = 0.32640$ cm^{-1}, the same table Assignment 1 Question 5 uses). The S_N sequence walks down onto it monotonically: 0.609042 at S_{10}, 0.606407 at S_{16}, 0.605631 at S_{24}, 0.605195 at S_{48}.

P_N against S_{N+1}

In slab geometry the two are algebraically the same interior equations. Expanding ψ in $N+1$ Legendre moments and truncating gives the same $N+1$ coupled ODEs as collocating at the $N+1$ Gauss–Legendre points, because a Gauss–Legendre rule of order $N+1$ integrates polynomials up to degree $2N+1$ exactly. **Any difference between P_N and S_{N+1} is therefore a difference of boundary condition**, and it is largest at low N where the boundary is the whole story:

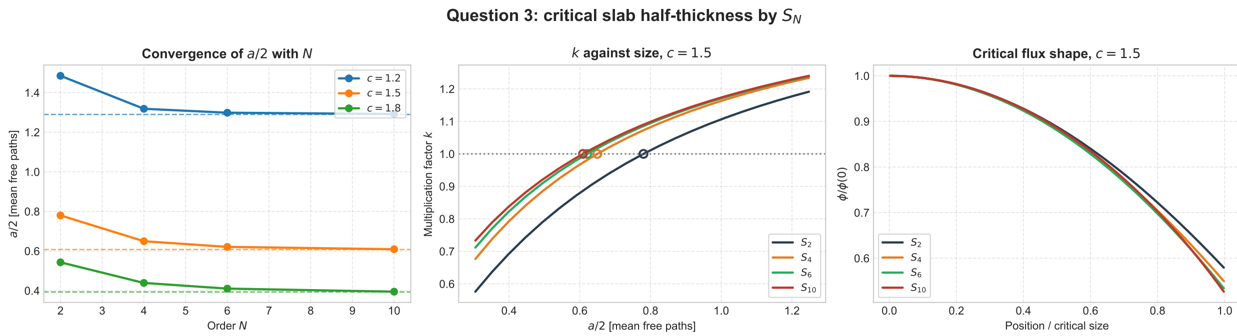

Figure 3: Left: convergence of $a/2$ with N, dashed lines the exact-transport values. Centre: k against half-thickness at $c = 1.5$, circles the $k = 1$ crossings. Right: the normalised critical flux, which barely moves with N — the order affects the boundary, not the interior.

- S_{N+1} sets $\psi(a/2, \mu_m) = 0$ on each incoming ordinate — the *Mark* condition.
- P_N needs conditions on moments instead. *Marshak*'s, $\int_0^1 P_{2j+1}(\mu)\psi\, d\mu = 0$, weight the whole incoming half-range; Mark's pin the same isolated directions S_{N+1} does.

With Mark conditions the two agree; with Marshak they do not. Concretely at $c = 1.5$: S_2 above gives $a/2 = 0.78001$, and P_1 with the Mark extrapolation $l_0 = 1/\sqrt{3}$ gives $\arctan(1/Bl_0)/B = 0.78001$ — the same five digits — while P_1 with the Marshak $l_0 = 2/3$ gives 0.72348, a 7% spread between two methods that share every interior equation.

A second source of difference is that S_N is a collocation method: it satisfies the equation exactly along N directions and says nothing between them. At the vacuum face the leaked current $\sum_{\mu_m>0} w_m \mu_m \psi_m$ is that of a flux strongly peaked towards $\mu = 1$, which a coarse quadrature underestimates — so the slab has to be made *thicker* than it should be to stay critical, and the table comes down from above.

All of that is a statement about *slabs.* In the sphere the equivalence fails at the interior equations themselves: the angular derivative $\frac{1-\mu^2}{r}\frac{\partial\psi}{\partial\mu}$ couples the Legendre moments in a different pattern from the way the α recursion of (29) couples the ordinates, so P_1 and S_2 are genuinely different approximations there rather than the same one under two boundary conditions. At $c = 1.5$, S_2 gives $R_c = 1.607$ mean free paths where P_1 with the Mark extrapolation gives 1.919 and with Marshak's 1.822 — a spread no choice of boundary condition can close.

Question 4: Critical sphere radius by S_N

Reflective at $r = 0$, vacuum at $r = R$, 100 cells, with the angular redistribution term of (29). The reference is Assignment 1's $\Sigma_t R_c = \pi\nu_0 - z_0$.

Cross-check between the two geometries. Both drive the asymptotic flux to zero at the same extrapolated surface, so $a/2 + z_0 = \frac{\pi}{2}\nu_0$ and $R_c + z_0 = \pi\nu_0$, and therefore

$$z_0 = R_c - 2\,(a/2) \tag{34}$$

which lets the two tables above check each other without any further calculation:

c	order	R_c [mfp]	exact transport [mfp]	departure
1.2	S_2	3.06060	3.17204	-3.51%
1.2	S_4	3.14631	3.17204	-0.81%
1.2	S_6	3.15991	3.17204	-0.38%
1.2	S_{10}	3.16728	3.17204	-0.15%
1.5	S_2	1.60725	1.69010	-4.90%
1.5	S_4	1.66739	1.69010	-1.34%
1.5	S_6	1.67934	1.69010	-0.64%
1.5	S_{10}	1.68595	1.69010	-0.25%
1.8	S_2	1.11753	1.18295	-5.53%
1.8	S_4	1.16413	1.18295	-1.59%
1.8	S_6	1.17412	1.18295	-0.75%
1.8	S_{10}	1.17971	1.18295	-0.27%

Table 9: Critical radius in mean free paths. Note the sign: the sphere converges from *below*, the slab from above.

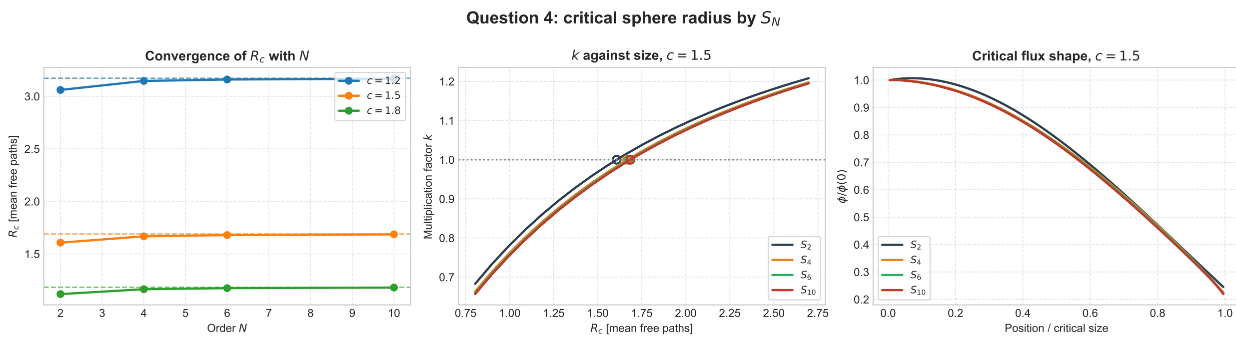

Figure 4: The same three panels for the sphere. The S_2 curve is the outlier in all three: in curvilinear geometry S_2 is not the P_1 equation, and the angular redistribution term it misrepresents is a loss from the interior directions.

c	from S_2	from S_4	from S_6	from S_{10}	Case's table
1.2	0.091	0.509	0.563	0.583	0.592
1.5	0.047	0.368	0.438	0.468	0.475
1.8	0.032	0.287	0.354	0.391	0.397

Table 10: Extrapolation distance z_0 implied by the two geometries, in mean free paths. At S_{10} it is 1.5% below Case's value at all three c — the same shortfall three times, which is the relation's own asymptotic error. At S_2 it nearly vanishes, which is the sharpest statement of what low-order S_N gets wrong: the boundary, not the interior.

Question 5: Critical mass of U-235 and Pu-239 by S_N

The cross sections are the Question 5 table of Assignment 1 — $\Sigma_t = 0.32640$ cm^{-1} for both, giving $c = 1.500$ for Pu-239 and $c = 1.300$ for U-235 — and the critical mass follows from the converged radius as $M_c = \frac{4}{3}\pi R_c^3 \rho$. Here $\Sigma_s > 0$, so the inner iteration does real work: about ten sweeps per outer, against one for the c-only problems above. The orders are S_2 to S_{16} rather than the S_2–S_{10} of the previous questions, since the mass goes as the cube of the radius and is the more demanding quantity.

material	order	c	R_c [cm]	M_c [kg]	asympt. diffusion R_c [cm]	departure
Pu-239	S_2	1.500	4.9242	7.852	5.1890	−5.10%
Pu-239	S_4	1.500	5.1084	8.767	5.1890	−1.55%
Pu-239	S_8	1.500	5.1586	9.028	5.1890	−0.59%
Pu-239	S_{16}	1.500	5.1730	9.104	5.1890	−0.31%
U-235	S_2	1.300	7.1218	28.748	7.4339	−4.20%
U-235	S_4	1.300	7.3514	31.619	7.4339	−1.11%
U-235	S_8	1.300	7.4072	32.345	7.4339	−0.36%
U-235	S_{16}	1.300	7.4231	32.553	7.4339	−0.15%

Table 11: Critical radius and mass, against the asymptotic diffusion result of Assignment 1 Question 5. The mass is the cube of the radius, so the 5% radius error of S_2 is a 14% error in mass.

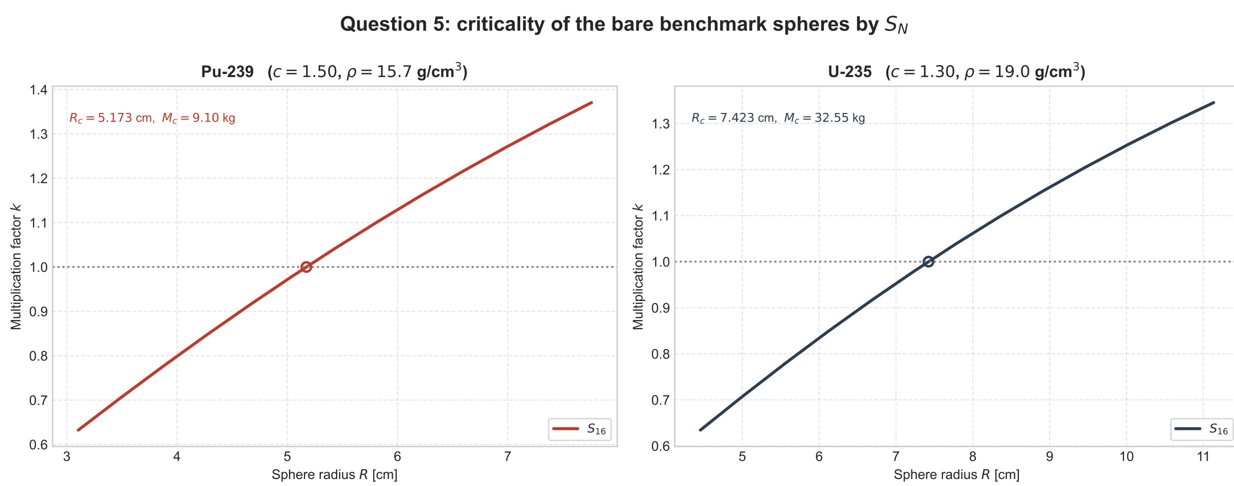

Figure 5: k against sphere radius at S_{16}, with the $k = 1$ crossing that fixes the critical mass.

Diffusion overestimates both radii, and by more for Pu-239 than for U-235: the surface sits about two mean free paths from the centre in both cases, but the higher c makes the Pu-239 flux fall off faster and its surface angular flux more forward-peaked, which is exactly what a diffusion approximation cannot represent. S_{16} recovers the difference to within 0.3% of the transport-theory radius.